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Anemia of chronic disease/anemia of inflammation

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INTRODUCTION

Anemia of chronic disease (ACD; also called anemia of inflammation [AI]) was initially thought to be associated primarily with infectious, inflammatory, or neoplastic disease. However, other observations have shown that ACD/AI can be seen in a variety of conditions including obesity, diabetes mellitus, congestive heart failure, chronic pulmonary diseases, critical illness and severe trauma, and other forms of diseases associated with acute or chronic immune activation.

The pathogenesis, laboratory findings, and treatment of ACD/AI will be reviewed here. Overviews of the evaluation of anemia are presented separately.

- Children – (See "Approach to the child with anemia".)
- Adults – (See "Diagnostic approach to anemia in adults".)

PATHOGENESIS

Reduced iron availability — ACD/AI is thought to result from an evolutionary defense strategy of the body to limit the availability of iron for invading microbes [1-3]. Accordingly, the pathophysiology of ACD/AI involves:

- Immune-mediated dysregulation of iron homeostasis via:
  - Hepcidin, the master regulator of iron homeostasis
  - Cytokines
- Immune-mediated effects on:

- Proliferation of erythroid progenitor cells
- Red blood cell (RBC) turnover and half-life
- Biologic activity of erythropoietin (EPO)

The increase in hepcidin (and other cytokines) causes iron to be retained within cells of the reticuloendothelial system, as illustrated in the figure (figure 1).

This in turn reduces the availability of iron for erythroid progenitor cells in the bone marrow. Thus, a hallmark of ACD/AI is the combination of low circulating iron levels, low transferrin saturation (TSAT), and normal or high concentration of the circulating iron storage protein ferritin [4]. (See 'Iron studies' below.)

The serum ferritin concentration is increased by tissue iron levels and inflammatory signals in ACD/AI, making it less useful as a measure of iron stores, unless it is below the normal range, which is good evidence of iron deficiency (see 'Iron studies' below). The mechanisms of the ferritin increase are multifactorial and include the effect of cytokines (ferritin serves as an acute phase reactant), destruction of tissues, macrophage iron accumulation, and active secretion of ferritin by macrophages [5,6]. (See 'Cytokine effects' below and "Acute phase reactants".)

Hepcidin (primary regulator of iron homeostasis) — Hepcidin is a small peptide mainly produced by the liver in response to cytokines or exposure to bacterial antigens, as a component of the innate immune response to infection [7,8].

Hepcidin binds to the plasma membrane channel ferroportin, blocking iron export [9]; this leads to internalization and degradation of ferroportin [8,10,11]. Removal of the ferroportin channel in turn prevents iron absorption in the small intestine, iron transport across the placenta, and iron release from macrophages [8].

Increased iron released from macrophages is the major source of iron for the heme synthesis required for erythropoiesis. Macrophages scavenge and ingest senescent RBCs; iron recycling from these RBCs accounts for 90 to 95 percent of the daily total body iron requirement [12]. Hepcidin also reduces the transfer of dietary iron or oral iron supplements from duodenal enterocytes into the circulation by the same actions on enterocyte ferroportin [13,14]. These changes are summarized in the figure (figure 1) and discussed in more detail separately. (See "Regulation of iron balance", section on 'Hepcidin'.)

Production of hepcidin by other cells, such as monocytes, has also been reported in inflammation; this is thought to form an autocrine mechanism to increase macrophage iron sequestration [15]. (See 'Reduced iron availability' above.)

In contrast, hepcidin is low in iron deficiency and iron deficiency anemia [16-18].

Hepcidin production, decreased urinary excretion of hepcidin, and increased serum hepcidin levels have been seen in patients with infection, malignancy, or an inflammatory state (as evidenced by C-reactive protein [CRP] levels >10 mg/dL) [13,19-23]. The complex role of hepcidin in different infections may go beyond iron homeostasis regulation [1]. As example:

- In certain skin infections such as necrotizing fasciitis caused by group A *Streptococcus*, the role of hepcidin in the skin is to induce expression of CXCL1 to recruit neutrophils [24].
- In hospitalized patients with COVID-19, increased hepcidin production and iron dyshomeostasis, indicated by changes in serum ferritin levels, has been linked to a poor clinical course and outcome [25-27].
- As an antimicrobial peptide, hepcidin produced by intestinal dendritic cells might protect the local mucosa by sequestering iron from the local microbiome, favoring mucosal healing in inflammatory bowel disease (IBD) [28].

Cytokine effects — Cytokines affect iron homeostasis by multiple mechanisms that promote iron storage in macrophages and reduce iron availability for RBC production (figure 1).

- **Increased hepcidin** – Hepcidin is the primary controller of iron availability to developing RBCs, as described above (see 'Hepcidin (primary regulator of iron homeostasis)' above). Cytokines including interleukin (IL)-1, IL-6, and IL-22 induce hepcidin production, as evidenced by numerous preclinical and clinical studies:
  - In a mouse model, knockout of IL-6 completely blunted the induction of hepcidin in response to inflammation [14].
  - In a series of 92 consecutive patients admitted for sepsis, hepcidin levels at admission were high, increased with the number of systemic inflammatory response syndrome (SIRS) criteria, and correlated with both IL-6 levels and the subsequent decrease in hemoglobin over the following days [29]. In a series of 150 patients with severe trauma, urinary hepcidin levels were extremely high on admission; hepcidin was positively correlated with the Injury Severity Score (ISS) and the duration of anemia, and negatively correlated with hypoxia [30].
  - In patients with ACD/AI, hepcidin mRNA levels in monocytes were significantly correlated with serum IL-6 concentrations [15,21].
  - Patients with inflammatory conditions who are treated with an anti-tumor necrosis factor (TNF) antibody or an anti-IL-6 antibody have reductions in inflammatory markers such as IL-6, hepcidin, and/or CRP, which correlates with improvement in anemia

[22,31,32].

- Emerging evidence suggests that vitamin D may suppress hepcidin [33,34]. Vitamin D deficiency and anemia sometimes coexist, and correction of vitamin D deficiency can improve anemia in a certain percentage of patients; this is believed to result from direct inhibition of hepcidin formation by the active vitamin [33,34].

The molecular mechanisms responsible for hepcidin induction continue to be elucidated; some of the proposed pathways are illustrated in the figure (figure 2) and discussed separately. (See "Regulation of iron balance", section on 'Hepcidin'.)

- **Increased macrophage iron uptake and retention** – Cytokines including IL-1, IL-6, IL-10, TNF, and interferon (IFN)-gamma stimulate the uptake of iron into macrophages by various pathways and promote intracellular iron storage by increasing the expression of the iron storage protein ferritin [35,36]. IFN-gamma also causes transcriptional inhibition of ferroportin expression, further contributing to macrophage iron retention [35].

  Inflammation-driven downregulation of ferroportin was determined to be due to Toll-like receptor-mediated induction of NFkB signaling and activation of histone deacetylases [37].

  The alterations of systemic iron trafficking and tissue iron distribution were visualized using R2* weighted magnetic resonance imaging; this indicated that in individuals with ACD/AI, iron is mainly retained in the spleen, which contains large proportions of macrophages, but not in the liver [38]. Iron content in the spleen increased with inflammatory activity and systemic ferritin levels.

- **Reduced erythropoietin** – Erythropoietin (EPO), the hormone produced by the kidney that promotes erythropoiesis, can be reduced by inhibitory cytokines, especially IL-1 beta, IFN-gamma, and TNF-alpha [36,39]. Expression and signaling via the EPO receptor on erythroid progenitor cells may also be impaired [40]. The lower EPO levels for a given degree of anemia contrast with most other anemic conditions such as iron deficiency anemia, in which serum EPO levels increase in proportion to the severity of anemia [20]. In addition, EPO receptor expression and EPO signaling in erythroid progenitor cells are impaired by the action of cytokines but also by iron deficiency, which reduces the expression of the EPO receptor component, Scribble [41,42]. Deletion of TFR2, the iron regulator that partners with the EPO receptor, increases EPO signaling independently of EPO levels and improves anemia in murine models of inflammation and chronic kidney disease (CKD) [43,44].

  Administration of EPO or other erythropoiesis-stimulating agents (ESAs) may have an antiinflammatory effect and reduce hepcidin expression [45-49]. Elevated levels of IL-6

and TNF-alpha correlated with a higher dose requirement for the ESA darbepoetin in individuals with kidney disease (see 'ESAs' below). In addition, EPO induces the expression of erythroferrone by bone marrow erythroblasts, and erythroferrone inhibits hepcidin production and increases iron availability for erythropoiesis [50].

- **Reduced RBC production** – Various inflammatory mediators directly inhibit erythroid cell differentiation and proliferation or even induce apoptosis via ceramide or radical mediated pathways [51-54]; increased apoptotic death of RBC precursors in the bone marrow has been observed in several studies [31].

  Cytokines such as IL-1 beta affect the balance between erythroid and myeloid cell production, activating the key myeloid transcription factor, PU.1 via NFkB signaling. Interferon gamma promotes myelopoiesis at the expense of erythropoiesis [55].

- **Shortened RBC survival** – Inflammatory cytokines may reduce RBC life span by driving free radical formation, with membrane lipid peroxidation and increased expression of macrophage scavenger receptors that facilitate ingestion of damaged RBCs (erythrophagocytosis) [36,53,56]. Increased macrophage activity is considered a minor contribution to the anemia in ACD/AI [36]. However, erythrophagocytosis and hemolysis may play a critical role in the initiation phase of anemia in patients with acute critical illness and sepsis [4,57].

  Erythrophagocytosis results from a combination of altered surface properties of RBCs, increased RBC fragility, and enhanced phagocytic activity of activated macrophages in the spleen, circulation, and other reticuloendothelial organs [58]. The maximal expression of this process is hemophagocytic lymphohistiocytosis (HLH)/macrophage activation syndrome.

  In an inflammatory mouse model, increased erythrophagocytosis in the spleen induced expression of the transcription factor SPI-C. Increased amounts of SPI-C, in association with TLR signaling, promoted the formation of stress erythroid progenitors (SEPs) in the spleen to compensate for anemia and hypoxia [59]. It remains to be shown whether this applies to humans.

EPIDEMIOLOGY

ACD/AI is considered the second most common cause of anemia worldwide, after iron deficiency anemia [36]. However, detailed statistics on its prevalence are not available. Often, the anemia in individuals with inflammatory diseases is complex and multifactorial, and it may be challenging to separate out the component due to ACD/AI. Examples of the prevalence of ACD/AI in various inflammatory states include the following:

- **Systemic inflammatory disorders** – Anemia is observed in 33 to 60 percent of patients with rheumatoid arthritis and in a variable percentage in patients with systemic lupus erythematosus, inflammatory bowel disease, vasculitis, or systemic sclerosis [60]. (See "Hematologic complications of rheumatoid arthritis", section on 'Anemia'.)

- **Infections** – ACD/AI is found in 18 to 95 percent of patients with infections; as with most disorders with inflammatory anemia, the prevalence of anemia is linked to disease severity [36]. Bacterial, viral, parasitic, and fungal infections may all be accompanied by ACD/AI. Among patients hospitalized with COVID-19, >40 percent presented with anemia upon admission, and the prevalence of anemia increased to >80 percent within one week [61].

- **Malignancy** – Cancer-related anemia occurs in more than 30 percent of the cases at diagnosis [62]; the rate reached 63 percent in one large study [63]. However, cancer-related anemia is multifactorial and includes causes such as deficiencies of iron and vitamins, bleeding and chemotherapy side effects, or bone marrow infiltration by malignant cells. Anemia is especially common in hematologic malignancies such as lymphoma and multiple myeloma. (See "Causes of anemia in patients with cancer".)

- **Aging** – ACD/AI accounts for approximately one-third of the cases of anemia of the older adult, often due to concomitant inflammatory conditions or chronic kidney disease [64]. (See 'Typical presentation' below.)

- **Chronic disorders** – Chronic disorders such as congestive heart failure, chronic obstructive pulmonary disease (COPD), and excess body weight can also cause ACD/AI [65]. Chronic kidney disease (CKD) is sometimes classified as a cause of ACD/AI and sometimes as a condition in the differential diagnosis. Some of the mechanisms are similar, but CKD is also characterized by EPO deficiency. (See 'Differential diagnosis' below.)

- **Anemia of critical illness** – Critically ill patients with massive acute inflammatory activity, such as individuals admitted to the intensive care unit (ICU) after major surgery, severe trauma, or sepsis, often show anemia of various degrees. Although anemia may be multifactorial, it has features of ACD/AI, with the further component of shortened RBC survival and erythrophagocytosis.

CLINICAL PRESENTATION

Typical presentation — The typical patient with ACD/AI has a known underlying chronic condition with an inflammatory component. While these were initially described as being infectious (eg, tuberculosis, pneumonia, endocarditis), inflammatory (eg, rheumatoid

arthritis, systemic lupus erythematosus [SLE], inflammatory bowel disease), or malignant (eg, lymphoma, solid organ cancer), several other diseases have been shown to have an inflammatory component and share some or all of the features of ACD/AI. (See 'Underlying disorders' below.)

In many cases, the anemia develops gradually and is detected on routine laboratory testing, and it is only on review of prior blood counts that the development of anemia is appreciated. The anemia is typically normochromic, normocytic, hypoproliferative (associated with a low or inappropriately low reticulocyte count), and mild to moderate in degree [4]. The white blood cell count and platelet count are unaffected by ACD/AI, although they may be abnormal secondary to the underlying disorder. (See 'Testing for all individuals' below.)

Likewise, the patients may be more symptomatic from their underlying condition than from their anemia, although fatigue may be common to both and may be challenging to attribute. However, anemia from ACD/AI and/or iron deficiency negatively impact the patient's quality of life, cardiovascular performance, exercise tolerance and mental fitness, similar to the effects of iron deficiency anemia [66].

As the chronic underlying conditions are more common with aging, ACD/AI is often seen in older adults. Anemia in older adults is often multifactorial, and ACD/AI may be seen together with other contributing factors such as chronic kidney disease, impaired nutrition, deficient iron intake or absorption, one or more vitamin deficiencies, hemodilution, and/or smoldering hematologic malignancies [64].

Underlying disorders — A large number of conditions with acute or chronic immune activation have been associated with ACD/AI, including infections, inflammatory disorders, malignancy, chronic obstructive pulmonary disease (COPD), acute and chronic heart failure, trauma, sepsis and critical illness, aging, high body mass index, or diabetes mellitus [4,36,67-69]. These are summarized in the table (table 1) and expanded in the following topic reviews:

- **Infections** – ACD/AI can frequently occur in many types of infections. The prevalence is mainly linked to disease severity and the associated degree of inflammation. In addition to ACD/AI, specific effects of the pathogen such as hemolysis, as exemplified by malaria, trypanosoma infection, or bone marrow infiltration, along with cytopathic effects toward erythroid progenitors (as in human immunodeficiency virus [HIV] infection) can also be of relevance. (See "HIV-associated cytopenias", section on 'General concepts'.)

- **Rheumatologic disorders** – These include rheumatoid arthritis, systemic lupus erythematosus, systemic sclerosis, seronegative arthritis, and vasculitis. (See "Systemic lupus erythematosus: Hematologic manifestations", section on 'Anemia of chronic

disease/anemia of inflammation'.)

- **Inflammatory bowel disease** – ACD/AI is especially common during acute exacerbations of inflammatory bowel disease; iron deficiency may often coexist. (See "Vitamin and mineral deficiencies in inflammatory bowel disease", section on 'Iron' and 'Concomitant iron deficiency' below.)

- **Castleman disease** – (See "HHV-8/KSHV-associated multicentric Castleman disease", section on 'Signs and symptoms' and "HHV-8-negative/idiopathic multicentric Castleman disease", section on 'Common signs, symptoms, and laboratory features'.)

- **Malignancy** – (See "Causes of anemia in patients with cancer" and "Role of ESAs in adults with non-hematologic cancers".)

- **Heart failure** – Inflammatory anemia is common in individuals with acute and chronic heart failure, with a higher prevalence in the former condition. However, the true prevalence varies according to the definition of iron deficiency applied and also involves additional causes such as hemodilution due to cardiac insufficiency [70]. (See "Evaluation and management of anemia and iron deficiency in adults with heart failure", section on 'Increased circulating cytokines and the anemia of inflammation'.)

- **Chronic obstructive pulmonary disease** – A subset of patients with COPD, estimated at approximately 50 percent, have laboratory findings consistent with ACD/AI (eg, anemia, elevated levels of C-reactive protein, IL-6, interferon-gamma, and decreased serum erythropoietin), suggesting the presence of background infection or inflammation [71,72]. (See "Chronic obstructive pulmonary disease: Diagnosis and staging".)

- **Pulmonary hypertension** – Individuals with pulmonary hypertension often have imbalances of iron homeostasis and ACD/AI, which impact the course of the disease [73]. (See "Clinical features and diagnosis of pulmonary hypertension of unclear etiology in adults".)

- **Obesity** – Many individuals with obesity present with signs of inflammation or immune activation that are linked to abnormalities of iron hemostasis and eventual development of anemia [74]. (See "Obesity in adults: Prevalence, screening, and evaluation".)

- **Diabetes and metformin use** – Type 2 diabetes mellitus may be associated with a higher prevalence of anemia, even in the absence of concomitant kidney disease [75,76]. However, little information is available on the underlying causes of anemia associated with type 2 diabetes. Some of the causes may be treatment related, given that metformin can cause vitamin B12 deficiency (other therapies such as sulfonamides do not [77]). (See "Causes and pathophysiology of vitamin B12 and folate deficiencies", section on 'Metformin'.)

- **Anemia of critical illness** – Acute event-related inflammatory anemia can occur within hours after surgery, major trauma, or sepsis. While this type of inflammatory anemia presents with the classical features of ACD/AI, a shortened half-life of red blood cells (RBCs), hemodilution, iatrogenic blood loss, and/or hemolysis may be of additional importance [36,78].

- **Kidney disease** – A generalized increase in the inflammatory response may occur in patients with decreased kidney function who are undergoing dialysis. In addition, the reduced urinary excretion of hepcidin can further contribute to macrophage iron retention. Of note, chronic kidney disease (CKD) causes anemia mostly as a consequence of decreased production of erythropoietin (EPO) in the kidney; however, in advanced CKD and in patients on dialysis, many typical features of ACD/AI are found that contribute to the pathogenesis and severity of anemia [79]. (See "Inflammation in patients with kidney function impairment", section on 'Inflammation and kidney disease' and 'Differential diagnosis' below.)

It is still a matter of debate whether (or to what extent) anemia itself contributes to the pathology and progression of inflammatory diseases or cancer, or if the poorer prognosis of individuals with inflammatory disorders plus anemia compared with inflammatory disorders alone is merely a reflection of a more advanced disease and an associated increase in inflammatory activity.

Concomitant iron deficiency — Some patients with ACD/AI may also have true iron deficiency, mainly as a consequence of acute or chronic bleeding episodes or concomitant conditions that affect iron absorption such as atrophic gastritis or celiac disease. (See "Determining the cause of iron deficiency in adolescents and adults", section on 'Causes (organized by mechanism)'.)

The proportion of patients who have both ACD/AI and iron deficiency is unclear and likely varies according to the patient population. Common examples include:

- Patients with inflammatory bowel disease (IBD) or colon cancer who have chronic bleeding along with inflammation.
- Menstruating females with a concomitant rheumatic disorder.
- Individuals living in low-resource countries where malaria is endemic and iron deficiency is often present due to nutritional deficits, chronic parasitic infections, or helminth infestations.
- Patients with CKD receiving hemodialysis, due to blood loss during the procedure.

Diagnostic phlebotomy for laboratory testing is rarely a cause of concomitant iron deficiency, although it may contribute to (or worsen) anemia, as individuals with ACD/AI may have inadequate mobilization of storage iron for RBC production. One possible

exception is individuals in the intensive care unit (ICU), who may have significant iron losses due to iatrogenic causes (frequent diagnostic testing, wasting blood from central line draws, hemodialysis).

There is no single biomarker that adequately distinguishes ACD/AI from iron deficiency in association with ACD/AI. In contrast, biomarkers can easily differentiate ACD/AI from true iron deficiency anemia. (See 'Testing for all individuals' below.)

Features suggestive of coexisting iron deficiency in an individual with ACD/AI include [4,13]:

- Microcytosis (see 'CBC, reticulocyte count, and blood smear review' below)
- Ferritin in the low or low-normal (rather than high-normal) range (see 'Iron studies' below)

Concomitant iron deficiency may complicate the interpretation of laboratory test results and may necessitate additional testing in some cases. Measurement of soluble transferrin receptor levels (sTfR) and/or the sTfR-ferritin index appears to be the most effective way to distinguish between ACD/AI and ACD/AI plus iron deficiency anemia [80]. Other tests such as the serum hepcidin concentration may also prove useful in making this distinction, especially when combined with results of the complete blood count (CBC) and other markers of iron balance. Other testing consistent with coexisting iron deficiency includes absent stainable iron on the bone marrow, and low reticulocyte hemoglobin concentration, although this testing is not always required. (See 'Additional studies in selected cases' below.)

If the distinction between ACD/AI and ACD/AI plus true iron deficiency cannot be made by laboratory tests alone, one may monitor the response to a short course of iron supplementation (a dose of intravenous iron or two weeks of oral iron); a response to iron supplementation suggests iron deficiency rather than ACD/AI. (See 'Iron supplementation' below.)

DIAGNOSTIC EVALUATION

Suspecting the diagnosis — ACD/AI is suspected in a patient with an acute or chronic infectious process, inflammatory disorder, or malignant condition who has mild to moderate normocytic, normochromic, hypoproliferative (ie, no evidence for an increased erythropoietic rate) anemia. (See 'Clinical presentation' above.)

In many cases, the underlying disorder leading to ACD/AI has already been diagnosed. While the list is not exhaustive, candidate conditions commonly leading to ACD/AI are listed above along with the appropriate UpToDate reviews dealing with these diagnoses. (See 'Underlying disorders' above.)

However, when alterations of iron homeostasis consistent with the diagnosis of ACD/AI have been obtained but the underlying disorder is unknown or not immediately apparent, further clinical and laboratory evaluation of the patient is required. This involves reviewing the patient's medical record for information concerning past diagnoses, timing of onset of the anemia, and whether age- and sex-appropriate cancer screening has been performed. A complete history and physical examination and routine laboratory testing for kidney and liver disease are also warranted.

Testing for all individuals

CBC, reticulocyte count, and blood smear review — The anemia in ACD/AI varies in severity. It is typically mild to moderate (hemoglobin concentration of 10 to 11 g/dL or 8 to 10 g/dL, respectively) [4]. The red blood cells (RBCs) are normocytic and normochromic in the majority of cases, with a normal mean corpuscular volume (MCV) and mean corpuscular hemoglobin concentration (MCHC).

The hemoglobin is <8 g/dL in approximately one-fifth of cases and the RBCs are microcytic and hypochromic in less than one-fourth of cases; these individuals may have concomitant iron deficiency [36,81]. In these cases the MCV is rarely less than 70 fL, the MCHC is normal to decreased, and the red cell distribution width (RDW) is normal to increased. (See 'Concomitant iron deficiency' above.)

The white blood cell and platelet counts are unaffected by ACD/AI, but they may be abnormal due to the underlying disorder (eg, leukocytosis in infection, leukopenia, and thrombocytopenia in systemic lupus erythematosus [SLE]).

The anemia is hypoproliferative, reflecting a decrease in RBC production due to insufficient EPO production and signaling along with limited iron availability, with a low (or inappropriately low) reticulocyte count. The absolute reticulocyte count is frequently <25,000/microL.

The blood smear shows a relatively uniform population of normal-sized RBCs without evidence of hemolysis (ie, no polychromatophilia, spherocytosis, or RBC fragments).

Findings related to the underlying disorder may be present on the peripheral smear, such as leukocytosis with a "left shift" in infection, the presence of leukemic or malignant cells, or leukopenia/lymphocytopenia in those with cancer or acute or chronic disorders involving the immune system. (See "Approach to the child with lymphocytosis or lymphocytopenia", section on 'Lymphocytopenia' and "Approach to the patient with neutrophilia", section on 'Peripheral blood smear' and "Evaluation of the peripheral blood smear", section on 'White blood cells'.)

Any findings that are **not** consistent with ACD/AI should be thoroughly evaluated so as

not to miss other causes of anemia that may require different treatments. This is especially important if anemia is severe, since in typical cases, the anemia of ACD/AI is mild to moderate. (See 'Tests to exclude other causes of anemia' below.)

Iron studies — Iron studies should be obtained to evaluate the possibility of iron deficiency.

The iron studies in ACD/AI show low circulating but sufficient storage iron, with the following findings typically seen [20,36]:

- Serum iron concentration is low (normal range, 60 to 150 mcg/dL [0.6 to 1.5 mg/L]; 11 to 27 microM/L).
- Transferrin (also measured as total iron binding capacity [TIBC]) is low (normal range, 300 to 360 mcg/dL [3 to 3.6 mg/L]; 54 to 64 microM/L).
- Transferrin saturation (TSAT) is low (<20 percent in approximately four-fifths of cases; normal range, 20 to 45 percent). It may be "pseudo-normal" if patients have very low transferrin concentrations (eg, <200 mcg/dL) [20].
- Ferritin is normal or increased; generally >100 mcg/L (normal ranges, 30 to 200 mcg/L [30 to 200 ng/mL] for women and 30 to 300 mcg/L [30 to 300 ng/mL] for men); in some countries the upper threshold may be up to 400 mcg/L. Ferritin can reach extremely high values (up to 10,000 mcg/L [10,000 ng/mL] and higher) in hemophagocytic lymphohistiocytosis (HLH) or macrophage activation syndrome (MAS) from the combination of severe inflammation and macrophage iron accumulation secondary to hemophagocytosis. (See "Clinical features and diagnosis of hemophagocytic lymphohistiocytosis", section on 'Clinical and laboratory'.)

Some findings are also characteristic of iron deficiency anemia, including low serum iron and low TSAT (table 2). (See 'Differential diagnosis' below.)

In contrast, unlike ACD/AI, in iron deficiency, transferrin is generally increased and ferritin is decreased (ferritin typically <30 mcg/L in absolute iron deficiency, <100 mcg/L in ACD/AI plus iron deficiency, and <200 mcg/L in ACD/AI and dialysis-dependent kidney disease plus iron deficiency) [20,36,66,82,83]. A ferritin level below these thresholds (or below the normal reference range) is good evidence of iron deficiency, but a ferritin level above 200 mcg/L cannot be used to exclude iron deficiency when a chronic inflammatory state is present. If results are inconclusive, the soluble transferrin receptor (sTfR) or sTfR-ferritin index can be used to identify concomitant iron deficiency in the setting of ACD/AI [36,80,84,85]. (See 'Soluble transferrin receptor and sTfR-ferritin index' below.)

The mechanisms of these changes are discussed above. (See 'Pathogenesis' above.)

CRP and/or ESR — The erythrocyte sedimentation rate (ESR) and/or C-reactive protein (CRP) increase as part of the acute phase response and can be used as a general measure of inflammatory disease activity.

Regulation of these proteins and their mechanism of increase is discussed separately. (See "Acute phase reactants", section on 'Laboratory evaluation'.)

Normal ranges are as follows:

- **CRP** – 3 to 5 mg/L in most individuals
- **ESR** – <20 mm/hour for men or <30 mm/hour for women

Tests to exclude other causes of anemia — In most cases, tests to eliminate other common, treatable causes of anemia will have been done prior to the consideration of ACD/AI as the only diagnosis. This testing should be reviewed and/or obtained if not done previously, with an individualized approach to deciding which tests can reasonably be omitted.

We generally exclude the following causes of anemia with the following testing:

- **Hemolysis** – Haptoglobin, lactate dehydrogenase (LDH), and/or bilirubin. (See "Diagnosis of hemolytic anemia in adults", section on 'Laboratory confirmation of hemolysis'.)

- **Decreased kidney function** – Serum creatinine and estimated glomerular filtration rate (eGFR). (See "Chronic kidney disease in children: Clinical manifestations, evaluation, and diagnosis" and "Definition and staging of chronic kidney disease in adults".)

- **Liver disease** – Serum transaminases and other liver function tests. (See "Overview of liver biochemical tests" and "Approach to the patient with abnormal liver tests".)

- **Vitamin deficiencies** – Vitamin B12, vitamin D, and folate levels. (See "Clinical manifestations and diagnosis of vitamin B12 and folate deficiencies", section on 'Serum vitamin B12 and folate levels'.)

  - Folate testing may be omitted in individuals with a varied healthy diet in countries with routine supplementation of grains and cereals or other foods.

  - Vitamin D levels are especially useful for older individuals or those with risk factors for deficiency such as limited sunlight exposure [33].

- **Endocrine disorders** – Hormone levels (eg, thyroid-stimulating hormone [TSH] and cortisol). (See "Diagnosis of and screening for hypothyroidism in nonpregnant adults"

and "Diagnosis of adrenal insufficiency in adults".)

- **Malignancy** – Age-appropriate cancer screening; evaluation of abnormal findings on history, physical examination, or laboratory testing; if applicable, assessment of associated clonal or genetic markers when systemic hematologic malignancies including myelodysplastic syndrome (MDS) are suspected. (See "Approach to the adult with lymphocytosis or lymphocytopenia" and "Overview of the myeloproliferative neoplasms" and "Clinical manifestations, diagnosis, and classification of myelodysplastic syndromes (MDS)".)

Additional studies in selected cases — The following testing may be helpful in identifying ACD/AI with concomitant iron deficiency.

Soluble transferrin receptor and sTfR-ferritin index

- **sTfR** – Soluble transferrin receptor (sTfR; also called circulating transferrin receptor or serum transferrin receptor) is the circulating protein derived from cleavage of the membrane transferrin receptor on erythroid precursor cells in the bone marrow. In iron deficiency, TfR density increases on cell membranes, and truncated sTfR increases in the serum. In contrast, in ACD/AI, cytokines suppress the expression of TfR. Thus, the concentration of sTfR varies in a similar pattern to transferrin (increased in iron deficiency and normal in ACD/AI) [35]. This subject is discussed in depth separately. (See "Diagnosis of iron deficiency and iron deficiency anemia in adults", section on 'Iron studies (list of available tests)'.)

  Measurement of sTfR (or the ratio of sTfR to the logarithm of ferritin) can help to distinguish between iron deficiency anemia and ACD/AI [86,87]. However, in most cases, ACD/AI can be easily differentiated from iron deficiency anemia by proper interpretation of standard serum iron parameters including iron, transferrin, transferrin saturation (TSAT), and ferritin [4,36]. (See 'Iron studies' above.)

  Thus, we reserve measurement of sTfR and/or the sTfR-ferritin ratio for cases in which there is a need to distinguish between pure ACD/AI versus ACD/AI and concomitant true iron deficiency [84]. (See 'Concomitant iron deficiency' above.)

- **sTfR-ferritin index** – The ratio of sTfR to ferritin is effective in distinguishing between iron deficiency anemia and ACD/AI because sTfR is increased in iron deficiency and normal or reduced in ACD/AI, while serum ferritin is decreased in iron deficiency and normal or increased in ACD/AI.

  The ratio of sTfR to ferritin can be calculated by dividing the sTfR (expressed as mg/L) by the ferritin (expressed as mcg/L), or the ratio of sTfR to the logarithm of the ferritin concentration may be calculated (referred to as the sTfR-ferritin index). The following

interpretations may be used [36,80]:

- An sTfR-ferritin index <1 suggests ACD/AI.
- An sTfR-ferritin index >2 suggests iron deficiency anemia (or iron deficiency anemia plus ACD/AI).

An example of the sTfR-ferritin index in patients with ACD/AI, iron deficiency anemia, or both is illustrated in the figure (figure 3).

Bone marrow — Bone marrow aspiration and biopsy are generally not indicated in the diagnosis of ACD/AI. However, this testing may be important for evaluating certain associated disorders and/or if the cause of an unexplained anemia remains unclear, especially if there is suspected hematologic malignancy, bone marrow infiltration, or infection impacting erythropoiesis (eg, HIV, leishmaniasis). (See 'Differential diagnosis' below.)

If the bone marrow aspirate is performed, iron staining is helpful to determine whether storage iron is present. ACD/AI is generally characterized by normal or increased storage iron in bone marrow macrophages (picture 1), reflecting reduced iron export from macrophages due to the action of hepcidin, along with decreased or absent iron in erythroid precursors (ie, decreased number of sideroblasts), representing reduced iron availability for erythropoiesis (see 'Pathogenesis' above) [88]. In contrast, the absence of storage iron in RBC precursors in the bone marrow is a gold standard test for iron deficiency.

The bone marrow should also be closely examined for the presence of dysplasia, as a myelodysplastic syndrome may also present with normochromic, normocytic anemia. Bone marrow findings in ACD/AI and conditions in the differential diagnosis of ACD/AI include the following:

- **ACD/AI** – Bone marrow macrophages contain normal to increased iron, while erythroid precursors show decreased to absent amounts of iron (ie, decreased to absent sideroblasts) (picture 1).

- **Iron deficiency** – Stainable iron is absent from both macrophages and erythroid precursors. (See "Diagnosis of iron deficiency and iron deficiency anemia in adults", section on 'Differential diagnosis'.)

- **Sideroblastic anemias** – The diagnostic hallmark of congenital or acquired sideroblastic anemia is the presence of ring sideroblasts on bone marrow examination (picture 2). The amount of iron in bone marrow macrophages is strikingly increased due to ineffective erythropoiesis. Single or multi-lineage dysplastic changes may be seen in patients with myelodysplastic syndromes (MDS).

Evaluation of sideroblastic anemias, including genetic testing, is presented separately. (See "Sideroblastic anemias: Diagnosis and management".)

- **MDS** – Inflammation is an important component in MDS, and ACD/AI is a component of anemia in all these conditions. Single or multi-lineage dysplastic changes with increased sideroblasts, including ring sideroblasts, and alteration of iron parameters are seen in patients with low-risk myelodysplasia (MDS-RS) (picture 3) due to pathogenic variants in the spliceosome gene *SF3B1*. (See "Sideroblastic anemias: Diagnosis and management", section on 'Diagnostic approach' and "Clinical manifestations, diagnosis, and classification of myelodysplastic syndromes (MDS)", section on 'Clinical presentation'.)

Hepcidin level — Hepcidin is the primary regulator of iron availability to developing RBCs and is increased in chronic inflammatory conditions including ACD/AI. (See 'Pathogenesis' above.)

Hepcidin levels are increased in ACD/AI and low in iron deficiency anemia. Importantly, the negative regulatory effect of iron deficiency appears to dominate over cytokine-mediated hepcidin induction [16,89]. Thus, individuals with both ACD/AI and true iron deficiency have significantly lower hepcidin levels than individuals with ACD/AI alone [13,17,18].

Assays to measure serum hepcidin have become available in some clinical laboratories [85]. However, assays differ in methods and reagents, and a universal reference standard is not available. The age- and sex-specific reference ranges for the testing laboratory should be used [90]. (See "Regulation of iron balance", section on 'Hepcidin'.)

Hepcidin testing is not routinely used in the diagnosis of ACD/AI. However, it may be helpful (if available) in selected cases, such as those in which iron deficiency and inflammation are both suspected, since iron deficiency and inflammation have contrasting effects on hepcidin expression. Results of hepcidin testing may be used alone or in combination with other tools [91].

Serum EPO — Erythropoietin (EPO) is the circulating hormone produced by the kidney that promotes erythropoiesis in response to tissue hypoxia. EPO levels are generally high in iron deficiency anemia and lower in ACD/AI with comparable degrees of anemia (table 2) [20]. However, the range of EPO levels in individuals with anemia is generally too broad for the test to be of diagnostic use in distinguishing iron deficiency anemia from ACD/AI.

We generally reserve measurement of the EPO concentration for individuals with ACD/AI who have symptomatic anemia and/or anemia that does not improve following

treatment of their underlying disorder, with or without iron supplementation. In such cases, a low EPO level suggests an ongoing inflammatory block or perhaps more severe kidney disease than originally thought, whereas a very high EPO level suggests insufficient erythropoiesis due to a bone marrow disorder such as myelodysplastic syndrome.

Making the diagnosis — There is no single test that will reliably make the diagnosis of ACD/AI. Rather, a pattern of abnormalities and the exclusion of other possible diagnoses serve to make this diagnosis. Major challenges include distinguishing ACD/AI from iron deficiency in individuals with an inflammatory condition and identifying additional factors contributing to ACD/AI [92].

ACD/AI is most likely when all (or most) of the following are present (see 'Testing for all individuals' above):

- Normochromic, normocytic anemia (hemoglobin mostly between 10 and 12 g/dL)
- Low reticulocyte count (or inappropriately low for the degree of anemia)
- Low serum iron (generally <60 mcg/dL)
- Normal to low serum transferrin (generally <300 mcg/dL)
- Low transferrin saturation (TSAT; generally <20 percent)
- Normal to increased serum ferritin (>100 mcg/L)
- Elevated CRP (generally >5 mg/L)

If the pattern of this testing is confusing or equivocal, additional studies may be helpful in confirming or excluding the diagnosis. (See 'Additional studies in selected cases' above.)

The response of anemia to treatment of the underlying disorder also supports the diagnosis of ACD/AI; the response to iron supplementation suggests a component of iron deficiency. (See 'Management' below.)

Other tests — Several markers have been studied for evaluating ACD/AI, distinguishing ACD/AI from iron deficiency anemia, or identifying patients with both forms of anemia, either alone or in combination with iron metabolism parameters [91,93,94]:

- Percentage of hypochromic RBCs
- Reticulocyte hemoglobin content (ret-He, CHr)
- RBC hemoglobin content

In two studies, ret-He <26 pg/cell was a stronger predictor of iron deficiency and iron deficiency anemia than routine iron studies [95]; thus, a ret-He of >25 pg/cell may help in distinguishing between iron deficiency and ACD/AI, although there is some overlap when iron deficiency is mild.

While some of these tests have shown promise in identifying true iron deficiency in patients with ACD/AI, in many cases, this testing depends greatly on the availability of a specific apparatus and standardized pre-analytical procedures. Moreover, most of these tests have never been prospectively studied to evaluate their true diagnostic potential for distinguishing ACD/AI from ACD/AI plus iron deficiency.

Additional information about the reticulocyte hemoglobin content and reticulocyte hemoglobin concentration is presented separately. (See "Automated complete blood count (CBC)", section on 'Reticulocytes'.)

DIFFERENTIAL DIAGNOSIS

The differential diagnosis of normochromic, normocytic (normal mean corpuscular volume [MCV]), hypoproliferative (low reticulocyte count) anemia is relatively broad. The following disorders (or combinations of disorders) may be considered (table 1):

- **Endocrine disorders** – Like ACD/AI, various endocrine disorders including hyperthyroidism, hypothyroidism, panhypopituitarism, primary and secondary hyperparathyroidism, and possibly vitamin D deficiency can cause anemia with normochromic, normocytic red blood cells (RBCs) and a low reticulocyte count and evidence of adequate iron stores. Unlike ACD/AI, in these disorders there will be accompanying clinical signs and symptoms of the specific disorder along with laboratory evidence of associated hormone deficiencies. (See "Chronic kidney disease (newly identified): Clinical presentation and diagnostic approach in adults" and "Diagnosis of and screening for hypothyroidism in nonpregnant adults" and "Diagnosis of hyperthyroidism" and "Clinical manifestations of hypopituitarism" and "Primary hyperparathyroidism: Diagnosis, differential diagnosis, and evaluation".)

- **Iron deficiency anemia** – Like ACD/AI, iron deficiency can cause anemia with a low reticulocyte count. Unlike ACD/AI, iron deficiency progresses to more severe anemia with increasingly microcytic RBCs. Unlike ACD/AI, iron deficiency is characterized by low serum ferritin and absent bone marrow iron; hepcidin, which is low in iron deficiency anemia, and bone marrow evaluation are rarely used in diagnostic testing for these conditions. Additional similarities and differences are summarized in the table (table 2). (See "Diagnosis of iron deficiency and iron deficiency anemia in adults" and "Microcytosis/ Microcytic anemia".)

  Iron deficiency and ACD/AI can coexist, and the finding of one cannot be used to exclude the other. Testing to evaluate combined ACD/AI and iron deficiency is discussed above. (See 'Concomitant iron deficiency' above.)

- **Sideroblastic anemias** – Sideroblastic anemias include a number of inherited and acquired disorders with ring sideroblasts in the bone marrow. While the rare and heterogeneous inherited forms are usually microcytic, acquired forms generally produce normocytic, normochromic RBCs with a low reticulocyte count; and some may have an abnormally high MCV. Unlike ACD/AI, sideroblastic anemias will demonstrate specific abnormalities upon laboratory evaluation or genetic testing, and ring sideroblasts will be present in the bone marrow. (See "Sideroblastic anemias: Diagnosis and management".)

- **Myelodysplastic syndrome** – Myelodysplastic syndromes (MDS) are a heterogenous group of clonal hematopoietic cell disorders with dysplasia in the bone marrow and one or more peripheral blood cytopenias. The RBCs may be macrocytic and hyperchromic or normocytic and normochromic, the reticulocyte count is low, and anemia often predominates, especially during the initial phases of MDS progression. Unlike ACD/AI, MDS has specific cytologic, molecular, and chromosomal findings, as discussed separately. (See "Clinical manifestations, diagnosis, and classification of myelodysplastic syndromes (MDS)".)

- **Infections** – Certain infections can cause anemia by direct effects on erythropoiesis in the bone marrow. Examples include HIV, leishmaniasis, trypanosomiasis, or miliary tuberculosis. Like ACD/AI, the RBCs may be normocytic and normochromic and the reticulocyte count is low. Some infections such as malaria are characterized by prevalent hemolysis, but features of ACD/AI may also be present. Unlike ACD/AI, anemia in some infections is characterized by different pathophysiology, findings of the specific organism, and the need for specific antimicrobial therapies. (See "HIV-associated cytopenias", section on 'Anemia' and "Visceral leishmaniasis: Clinical manifestations and diagnosis" and "Anemia in malaria".)

- **Chronic kidney disease** – As noted above, chronic kidney disease (CKD) may have an inflammatory component and be a cause of ACD/AI. However, unlike ACD/AI, CKD also is characterized by deficiency of erythropoietin (EPO). (See 'Underlying disorders' above.)

MANAGEMENT

Overview of management — The following principles apply to the treatment of ACD/AI:

- The preferred initial therapy is treatment of the underlying disorder. (See 'Treatment of the underlying disorder' below.)

- Other causes of anemia should be identified and treated if possible. (See 'Tests to exclude other causes of anemia' above.)

- Iron supplementation is generally reserved for those with concomitant iron deficiency,

and erythropoiesis-stimulating agents (ESAs) are generally not used, with a few exceptions. (See 'Iron supplementation' below and 'ESAs' below.)

- Red blood cell (RBC) transfusions should be reserved for those with severe, life-threatening, and symptomatic anemia. (See 'Transfusion' below.)

The overall goal of therapy is to reduce symptoms and improve clinical outcomes, not necessarily to normalize the hemoglobin level. This approach is informed by the following observations:

- For many patients with ACD/AI, the anemia is mild, and their symptoms mainly relate to their underlying disease rather than anemia. However, this issue has not been systematically studied.

- There is only limited information on the impact of ACD/AI development on the course of the disease underlying ACD/AI. Of note, ACD/AI originates from the host response to inhibit the growth of iron-requiring microorganisms [96-98]. Accordingly, mild iron deficiency or anemia in environments with a high endemic burden of malaria have been shown to be protective in some cases [99]. Many studies have demonstrated a correlation between anemia and clinical outcomes, but a causative role for anemia in these disorders (or improved outcomes with anemia treatment) has been more challenging to establish [100,101].

- Studies in individuals with end-stage kidney disease have demonstrated increased mortality when hemoglobin levels were normalized by ESA treatment. Therapeutic endpoints for ACD/AI, extrapolated from these observations, recommend a target hemoglobin of 11 to 12 g/dL for individuals with ACD/AI and persistent underlying inflammatory disease [36,102].

Treatment of the underlying disorder — Typically, the underlying disorder(s) responsible for ACD/AI should be identified and properly treated.

The degree to which ACD/AI responds to treatment of the underlying disorder may depend on several factors, including whether the inflammatory component is controlled and the presence of other contributing factors. As an example, in patients with diabetes, improved glucose control may not lead to resolution of ACD/AI because it may not correct concomitant kidney disease or inflammatory changes caused by obesity.

In other cases, treatment of the underlying disorder may be more effective in improving the anemia. As examples:

- If the anemia is due to an underlying malignancy, it may be transiently exacerbated by the myelosuppressive effects of chemotherapy and/or radiation. In the long term,

successful cancer treatment may lead to anemia improvement. Treatment of cancer-associated anemia is discussed in depth separately. (See "Role of ESAs in adults with non-hematologic cancers".)

- If the anemia is due to an underlying disorder with a major inflammatory component (eg, rheumatoid arthritis, Castleman disease, vasculitis), treatment of the inflammatory disorder with a disease-modifying antirheumatic drug (DMARD) may lead to improvement in the anemia [60]. (See "Initial pharmacologic management of rheumatoid arthritis in adults", section on 'Approach to initial therapy' and "HHV-8-negative/idiopathic multicentric Castleman disease", section on 'IL-6 inhibitor-based therapy'.)

- If anemia is due to an underlying infection such as pneumonia, endocarditis, tuberculosis, or HIV infection, specific treatment or control of the infection will also result in resolution of the anemia, provided the anemia is solely the consequence of ACD/AI.

Iron supplementation — Altered iron trafficking with reduced iron availability to developing red blood cells (RBCs) is central to the pathogenesis of ACD/AI. (See 'Reduced iron availability' above.)

However, the use of iron supplementation in ACD/AI requires a careful estimation of benefits and risks for the individual.

- **People with concomitant iron deficiency** – Iron supplementation is the first-line treatment for individuals with both ACD/AI and true concomitant iron deficiency (documented or strongly suspected); these individuals require iron for normal physiologic function [46,47,66]. Iron treatment in that setting has been shown to be effective to induce a hematologic response and to replenish iron stores specifically in patients with low inflammatory activity [103-105]. Thresholds to identify iron deficiency depend on the patient's clinical status; a ferritin <100 mcg/L and a transferrin saturation (TSAT) <20 percent are accepted as indicators of iron deficiency in individuals with ACD/AI. (See 'Iron studies' above and "Diagnosis of iron deficiency and iron deficiency anemia in adults", section on 'Diagnostic evaluation'.)

  Oral and intravenous iron are both used for treating iron deficiency in individuals with ACD/AI, although there is a paucity of data from randomized trials to address the question of their efficacy [46,47]. Oral iron is most likely to be effective if hepcidin levels are normal or reduced, which is often the case in truly iron-deficient individuals [13]. In the other cases, intravenous iron may be more effective than oral iron. Acute flares of inflammatory bowel disease (IBD) are a classic example [45]. If an individual with iron deficiency does not have a hemoglobin response to oral iron, intravenous iron should be administered before considering the treatment ineffective. (See "Treatment of iron deficiency and iron deficiency anemia in adults", section on 'Approaches to lack of

response'.)

Response to iron treatment may well not be as robust as that of patients with iron deficiency anemia alone, until the underlying cause of the ACD/AI is improved by specific treatment. In a clinical trial of patients with ACD/AI due to tuberculosis, iron absorption (measured with stable isotopes) occurred only after successful tuberculosis treatment, suggesting that iron supplementation should be reserved for patients who remain anemic after specific treatment of the underlying disease [106].

These observations are consistent with a comparative analysis of iron therapy routes in preclinical models of inflammatory arthritis, which demonstrated that oral iron is poorly absorbed in individuals with ACD/AI, whereas intravenous iron is retained in macrophages of the liver and spleen and unavailable for erythropoiesis [107]. In contrast, both treatments were effective in ameliorating anemia when ACD/AI was associated with true iron deficiency along with ongoing inflammation. Thus, it is important to identify concomitant true iron deficiency when present in individuals with ACD/AI for guiding therapy and identifying sources of ongoing blood loss that may be depleting iron stores, as well as dietary habits or concomitant medications such as proton pump inhibitors that may be interfering with iron intake or absorption, respectively [108]. The choice of route, formulation, dose, and adverse effects of iron supplementation are discussed separately. (See "Treatment of iron deficiency and iron deficiency anemia in adults", section on 'Iron replacement products'.)

- **People without iron deficiency** – We generally do **not** give iron to individuals with ACD/AI who are iron replete.

- **Areas of uncertainty** – Uncertainties exist in regard to iron supplementation in patients with acute infection. We generally avoid iron supplementation in individuals with active infections due to the growth-promoting activity of iron for many pathogens and possible effects on immune function [2,3]. Supplementation of iron in areas with a high endemic burden of infection results in increased infection-related morbidity and mortality [109-111]. Debate is ongoing about whether iron supplementation in chronically ill patients such as those undergoing dialysis will increase the risk of infection.

  Similar uncertainties exist regarding the effects of iron supplementation in individuals with cancer. In a 2023 trial that investigated the effects of oral or intravenous iron given two weeks prior to surgery for colorectal cancer, less than 20 percent of patients in both groups had normalization of hemoglobin levels at the time of surgery; however, a higher proportion of individuals in the intravenous iron group had normalization of hemoglobin levels after longer follow-up [112]. The trial was limited by a lack of information on the causes of anemia, lack of data on the effect of iron therapy on the long-term outcome of

cancer, and a dosing schedule for oral iron (three times per day) that reduces iron absorption. As a result, a definitive recommendation regarding iron supplementation in cancer cannot be provided based on this trial [113,114]. Ongoing studies might help to clarify the issue of iron supplementation in cancer [115].

In some cases, addition of an erythropoiesis stimulating agent (ESA) to iron supplementation may be helpful, as ESAs may have anti-inflammatory effects and decrease hepcidin, both of which improve iron trafficking to developing RBCs [45,60,102]. (See 'Pathogenesis' above and 'ESAs' below.)

ESAs — Recombinant ESAs such as erythropoietin have been used for treatment of ACD/AI for many years [116-118].

However, concerns have arisen from studies showing higher mortality rates with the use of these agents in certain settings, including individuals with cancer, individuals undergoing dialysis who do not have a rapid hemoglobin response, or non-dialysis patients treated with newer erythropoiesis-stimulating drugs [119,120]. This resulted in recommendations to reduce ESA doses and to aim at target hemoglobin levels between 10 to 12 g/dL in patients with chronic kidney disease (CKD) receiving ESAs [121].

We generally do not use ESAs in individuals with ACD/AI, with the following exceptions in which an ESA may be appropriate:

- Individuals with CKD who may have a deficiency of erythropoietin (EPO). (See "Treatment of anemia in nondialysis chronic kidney disease" and "Treatment of anemia in patients on dialysis", section on 'ESAs'.)

- Selected individuals with cancer who are receiving chemotherapy. (See "Role of ESAs in adults with non-hematologic cancers".)

- Selected patients with low-risk myelodysplastic syndromes when serum EPO levels are low [122]. It is possible that such treatment will be replaced in the near future by the transforming growth factor beta (TGF-beta) superfamily ligand luspatercept. Indeed, a phase 3 randomized controlled clinical trial of this activin ligand trap versus ESA in low-risk transfusion-dependent MDS has shown a better response in terms of blood transfusion saving in the former group [123]. (See "Treatment of lower-risk myelodysplastic syndromes/neoplasms (MDS)".)

- Certain individuals with inflammatory bowel disease or rheumatologic disorders who do not have an adequate improvement in hemoglobin with iron supplementation [60,124]. (See "Treatment of iron deficiency and iron deficiency anemia in adults", section on 'Inflammatory bowel disease and iron-restricted erythropoiesis'.)

- Selected individuals scheduled for elective surgery. (See "Perioperative blood management: Strategies to minimize transfusions", section on 'Use of erythropoietin'.)

When an ESA is used, concomitant iron supplementation can reduce the doses of ESA to achieve a hematologic response, as clearly shown in CKD [125]. Supplemental iron should be administered, as needed, to maintain a TSAT of ≥20 percent and a serum ferritin level of ≥100 mcg/L [126]. (See 'Iron supplementation' above.)

Use of an ESA to treat ACD/AI, except in CKD or MDS, may be considered "unlabeled or investigational" in the United States and may not be reimbursed.

Transfusion — Transfusion of packed RBCs is appropriate when the patient with ACD/AI develops severe, life-threatening symptomatic anemia, and the clinician believes that there is insufficient time to wait for the anemia to improve with treatment of the underlying condition, iron administration, and/or an ESA. The use of transfusion in such settings as well as the trigger hemoglobin level for such treatment are discussed separately. (See "Indications and hemoglobin thresholds for RBC transfusion in adults", section on 'Overview of our approach'.)

Emerging therapies — A number of approaches to targeting the mediators of ACD/AI are under investigation [92]; some compounds target the hepcidin/ferroportin axis and others the hypoxia-related pathways [65]. These therapies must be evaluated clinically to assess whether they may overcome macrophage iron retention and/or immune-mediated inhibition of erythropoiesis, and their possible role must be evaluated in the different disorders characterized by ACD/AI [65].

- **Hepcidin and ferroportin** – Approaches with agents that alter/inhibit the function of hepcidin (eg, hepcidin antagonists) and increase the iron export activity of the hepcidin receptor (ferroportin) are under study to alleviate the various disorders of iron metabolism associated with increased levels of hepcidin, including ACD/AI [8,127]. (See 'Hepcidin (primary regulator of iron homeostasis)' above.)

- **Prolyl hydroxylase inhibitors** – Prolyl hydroxylase inhibitors (PHI) stabilize hypoxia-inducible factor (HIF), promote production of endogenous EPO, and increase intestinal iron absorption. Their use and efficacy in anemia associated with CKD is discussed separately. (See "Treatment of anemia in nondialysis chronic kidney disease", section on 'Treatment' and "Treatment of anemia in patients on dialysis", section on 'Treatment'.)

- **Other approaches** – Several new treatments interfering with bone morphogenetic protein (BMP) receptor/SMAD/TGF-beta signaling have shown promising results in preclinical models and may become applicable for ACD/AI treatment in the future [128]. Examples include luspatercept, momelotinib, and anti-hemojuvelin antibodies. The

applicability of these drugs for treating ACD/AI remains to be demonstrated.

SOCIETY GUIDELINE LINKS

Links to society and government-sponsored guidelines from selected countries and regions around the world are provided separately. (See "Society guideline links: Anemia in adults".)

INFORMATION FOR PATIENTS

UpToDate offers two types of patient education materials, "The Basics" and "Beyond the Basics." The Basics patient education pieces are written in plain language, at the 5th to 6th grade reading level, and they answer the four or five key questions a patient might have about a given condition. These articles are best for patients who want a general overview and who prefer short, easy-to-read materials. Beyond the Basics patient education pieces are longer, more sophisticated, and more detailed. These articles are written at the 10th to 12th grade reading level and are best for patients who want in-depth information and are comfortable with some medical jargon.

Here are the patient education articles that are relevant to this topic. We encourage you to print or e-mail these topics to your patients. (You can also locate patient education articles on a variety of subjects by searching on "patient info" and the keyword(s) of interest.)

- Basics topics (see "Patient education: Anemia of inflammation (anemia of chronic disease) (The Basics)")

SUMMARY AND RECOMMENDATIONS

- **Definition and prevalence** – Anemia of chronic disease/anemia of inflammation (ACD/AI) occurs when an acute or chronic inflammatory condition leads to increased cytokine production and elevated hepcidin, which causes iron to be sequestered in reticuloendothelial macrophages, making it unavailable to developing red blood cells (RBCs). After iron deficiency, ACD/AI is the second most common cause of anemia worldwide. (See 'Pathogenesis' above and 'Epidemiology' above.)

- **Presentation** – ACD/AI typically presents as mild to moderate anemia with normochromic, normocytic RBCs and a low reticulocyte count in an individual with a known chronic infectious or inflammatory disorder. Common underlying disorders are summarized in the table (table 1). Some individuals may have concomitant iron deficiency. (See 'Clinical presentation' above.)

- **Evaluation** – All individuals with suspected ACD/AI should have a complete blood count (CBC), reticulocyte count, review of the RBC indices or blood smear, and iron studies. A test for inflammation (typically C-reactive protein [CRP]) may be helpful, along with testing for hemolysis, kidney and liver function, hormone levels, and vitamin deficiencies. ACD/AI is characterized by low serum iron, low transferrin saturation, normal to increased serum ferritin, and evidence of an inflammatory state (table 2). Additional testing may be helpful in selected cases, especially those with possible concomitant iron deficiency. The diagnosis is generally based on the pattern of findings and exclusion of other anemias; there is no specific diagnostic test. (See 'Diagnostic evaluation' above.)

- **Differential** – The differential diagnosis includes other hypoproliferative anemias such as deficiencies of vitamins needed for hematopoiesis, endocrine disorders, myelodysplasia, hematologic malignancy, or chronic kidney disease (CKD) (table 1). (See 'Differential diagnosis' above.)

  Separate topics discuss the general anemia evaluation. (See "Approach to the child with anemia" and "Diagnostic approach to anemia in adults".)

- **Management** – The goal of treatment is to reduce symptoms and improve clinical outcomes, not to normalize the hemoglobin.

  - **Treat underlying disorder** – Treatment of the underlying disorder is paramount. Other causes of anemia should be identified and treated if possible. (See 'Overview of management' above and 'Treatment of the underlying disorder' above and 'Tests to exclude other causes of anemia' above.)

  - **Iron** – Iron supplementation is generally reserved for those with concomitant iron deficiency (ferritin <100 mcg/L and transferrin saturation [TSAT] <20 percent; elevated soluble transferrin receptor [sTfR] if standard iron studies are inconclusive). (See 'Iron supplementation' above.)

  - **ESAs** – Erythropoiesis-stimulating agents (ESAs) are generally reserved for selected patients with CKD or inflammatory bowel disease-associated ACD/AI with symptomatic anemia not responding to therapy for the underlying disease or ACD/AI and true iron deficiency not responding to iron supplementation. (See 'ESAs' above.)

  - **Transfusion** – RBC transfusions should be reserved for those with severe, life-threatening, and symptomatic anemia for which it is not possible to wait for a response to other therapies. (See 'Transfusion' above.)

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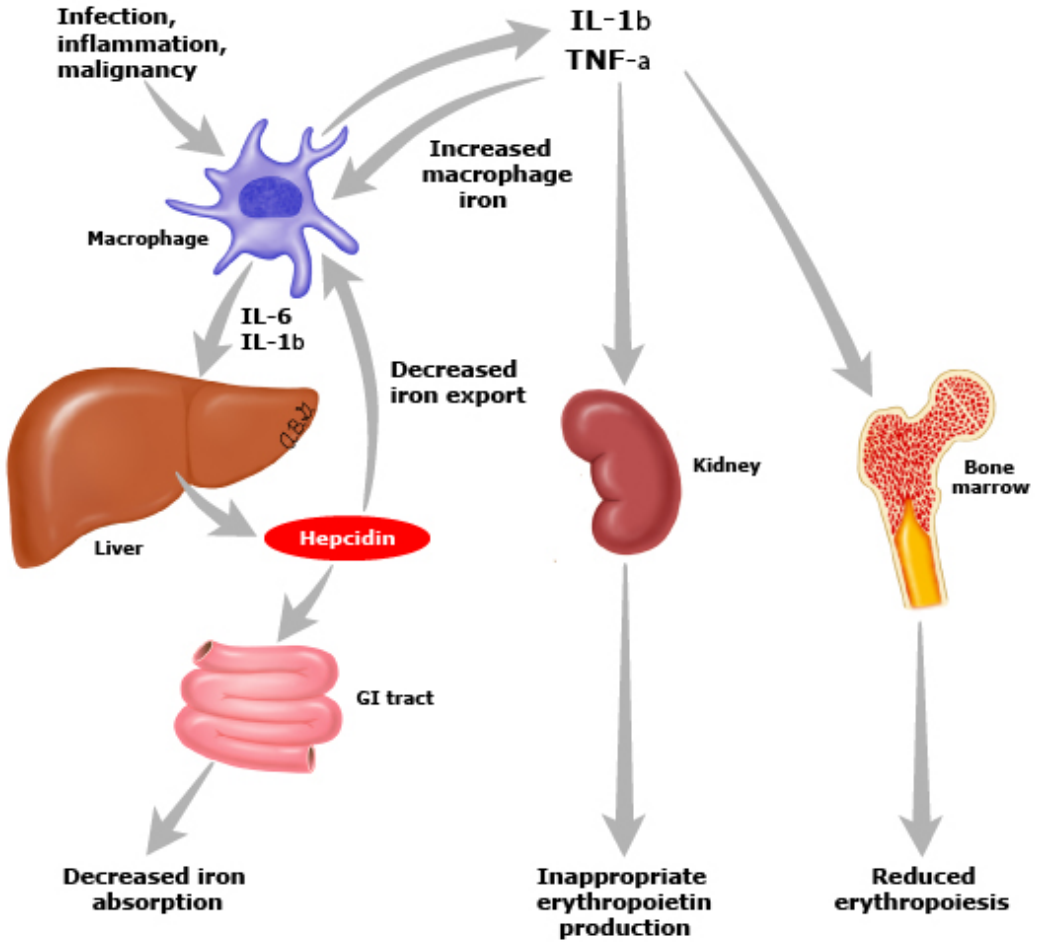
Topic 7149 Version 63.0

GRAPHICS

Figure 1: **Mechanism for anemia of chronic disease/anemia of inflammation (ACD/AI)**

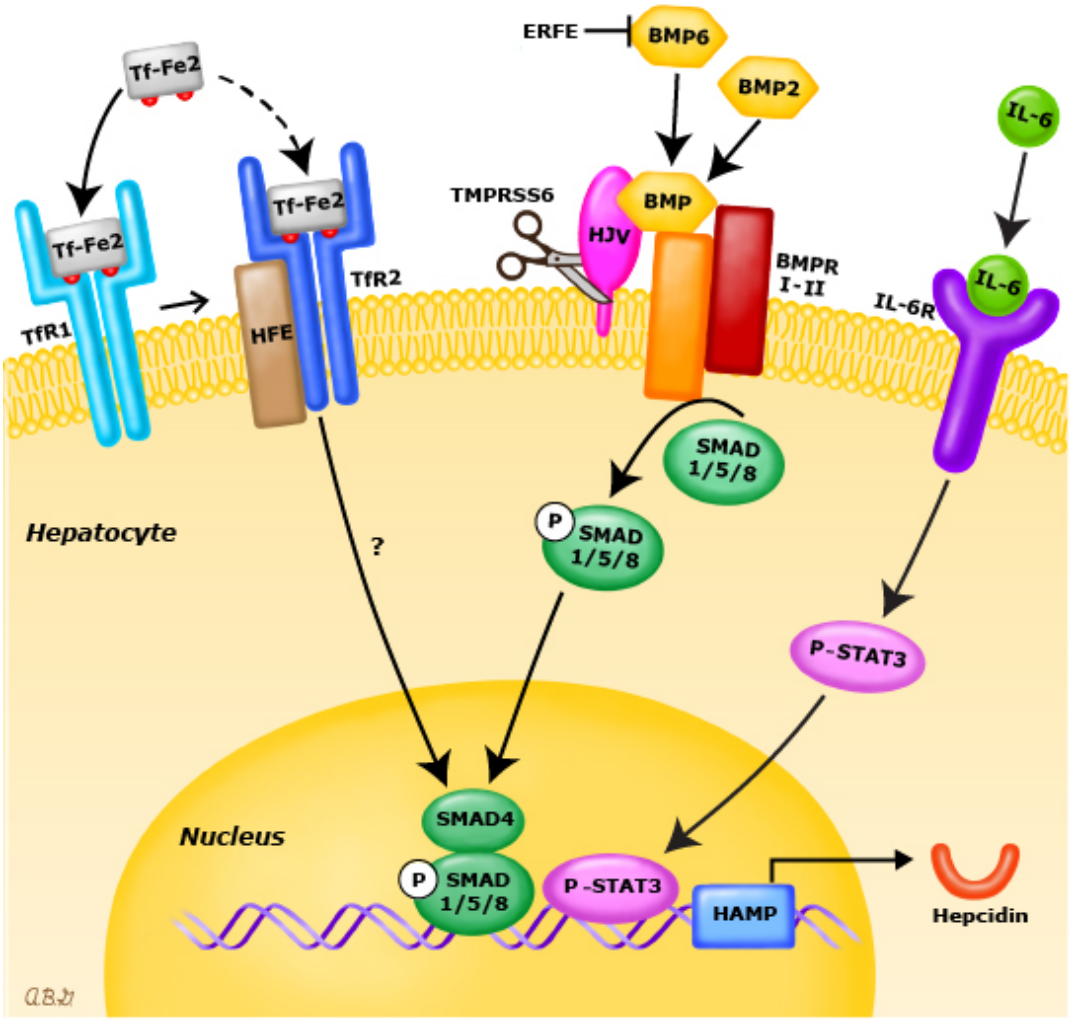


A proposed mechanism for ACD/AI is shown here. In the presence of infection, inflammation, or malignancy, the macrophage is stimulated to produce IL-6 and IL-1b, which induce the production of hepcidin by the liver. Hepcidin, in turn, through its interaction with the iron export protein ferroportin, inhibits iron absorption from the gastrointestinal tract and decreases release of iron from macrophages. Both effects lead to the reduced plasma iron levels (hypoferremia) characteristic of ACD/AI. Inflammatory cytokines such as IL-1b and TNF-a reduce erythropoietin production, decrease red blood cell lifespan, and reduce the efficiency of erythropoiesis, which are also components of ACD/AI.

ACD/AI: anemia of chronic disease/anemia of inflammation; GI: gastrointestinal; IL-1b: interleukin-1 beta; IL-16: interleukin-6; TNF-a: tumor necrosis factor-alpha.

Graphic 78381 Version 4.0

Figure 2: **Hepcidin regulatory pathways**



Schematic representation of mechanisms activating hepcidin transcription.

Central part – BMP6 and BMP2 activate type I and II receptors, interacting with the co-receptor hemojuvelin on hepatocyte plasma membrane. This triggers the phosphorylation of SMAD1/5/8, which interacts with SMAD4; the transcriptional SMAD complex translocates to the nucleus to activate transcription of the *HAMP* gene, which encodes hepcidin. ERFE can sequester BMP6 and other BMPs.

Left side (potential role of HFE and TfR2) – HFE binds TfR1 in competition with Tf-Fe^2. High Tf-Fe^2 levels displace HFE, which is then free to bind TfR2. It is hypothesized that the HFE-TfR2 complex activates hepcidin, but the molecular mechanisms are unclear (indicated by the question mark).

Right side – IL-6, increased in inflammation, interacts with its receptor (IL-6R) to phosphorylate STAT3, which in turn interacts with the HAMP promoter to activate hepcidin transcription.

Refer to UpToDate for further discussions of iron balance and information about specific proteins.

TfR1: transferrin receptor 1; Tf-Fe2: diferric transferrin; HFE: product of the hereditary hemochromatosis gene; HJV: hemojuvelin; TMPRSS6: matriptase-2 (protease); BMP6: bone morphogenetic protein 6; ERFE: erythroferrone; BMP2: bone morphogenetic protein 2; BMPRI-II: bone morphogenetic protein type I and II receptors; IL-6: interleukin 6; IL-6R: IL-6 receptor; HAMP: hepcidin antimicrobial peptide.

Graphic 64022 Version 6.0

Table 1: **Causes of anemia of chronic disease/anemia of inflammation (ACD/AI)**

Causes of ACD/AI – Inflammatory conditions in which cytokine production leads to altered iron trafficking and decreased production of RBCs.

Anemia is expected to improve with treatment of the underlying disorder.

- Aging (partial effect)
- Autoimmune and autoinflammatory diseases
  - Castleman disease
  - Inflammatory bowel disease
  - Rheumatoid arthritis
  - Sarcoidosis
  - SLE
  - Vasculitis
- Congestive heart failure
- COPD and pulmonary arterial hypertension
- Chronic kidney disease (advanced)
- Critical illness or trauma
- Hematologic malignancies and other cancers
- Infection
  - Complicated urinary tract, skin, or skin structure infections
  - Endocarditis
  - HIV infection
  - Other systemic bacterial, parasitic, viral, and fungal infections
  - Osteomyelitis
  - Pneumonia
  - Septicemia
  - Tuberculosis
- Obesity

The lists provide common examples (listed alphabetically) but are not exhaustive. Refer to UpToDate for details.

ACD: anemia of chronic disease; AI: anemia of inflammation; RBC: red blood cell; SLE: systemic lupus erythematosus; COPD: chronic obstructive pulmonary disease.

Graphic 126679 Version 3.0

Table 2: **Laboratory findings in iron deficiency anemia, alpha or beta thalassemia, and anemia of chronic disease/anemia of inflammation (ACD/AI)**

Test	Iron deficiency anemia	Alpha or beta thalassemia	ACD/AI
Complete blood count			
Hemoglobin	Decreased	Normal or decreased	Decreased
Mean corpuscular volume (MCV)	Normal to decreased	Decreased	Normal to decreased
Mean corpuscular hemoglobin concentration (MCHC)	Normal to decreased	Decreased	Normal to decreased
Red cell distribution width (RDW)	Increased	Mildly increased or normal	Normal to increased
RBC count	Decreased	Increased or normal	Decreased
Peripheral blood smear/RBC morphology	Anisocytosis, hypochromia	Anisopoikilocytosis, hypochromia	Normal
Iron studies			
Serum iron	Decreased	Normal or increased	Decreased
Total iron-binding capacity (TIBC); transferrin	Increased	Normal	Decreased
Transferrin saturation (TSAT)	Decreased	Normal	Normal to decreased
Serum ferritin	Decreased	Normal or increased	Normal to increased
Erythrocyte protoporphyrin*	Increased	Normal or increased	Increased
Soluble transferrin receptor (sTfR)*	Increased	Increased	Normal
Reticulocyte hemoglobin equivalent (Ret-he or CHr)	Decreased	Decreased	Normal
C-reactive protein	Normal	Normal	Increased
Serum hepcidin*	Decreased	Normal to decreased	Increased
Serum erythropoietin*	Increased	Increased	Normal to moderately increased

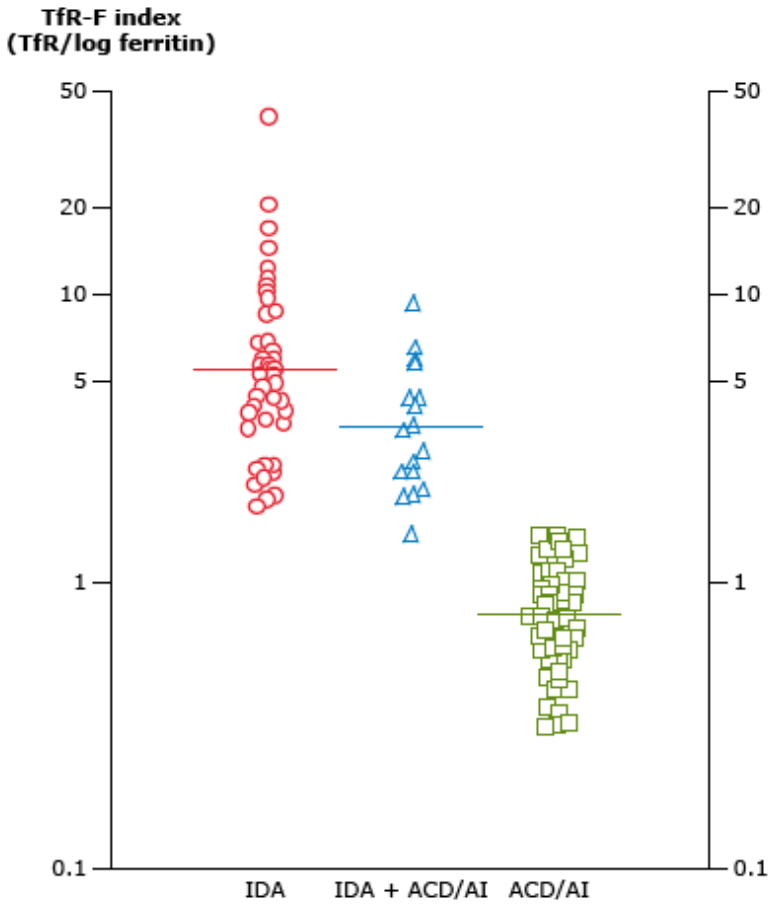
Refer to UpToDate topics on anemia for further details of the evaluation and interpretation. Refer to UpToDate topics on iron deficiency anemia for early changes associated with iron deficiency (before anemia develops).

ACD/AI: anemia of chronic disease/anemia of inflammation; RBC: red blood cell.

* Not used in the routine evaluation of anemia. Erythropoietin is not useful unless the hemoglobin is <10 g/dL.

Graphic 65701 Version 11.0

Figure 3: **Serum transferrin receptor-ferritin index in anemic patients with iron deficiency, anemia of chronic disease/anemia of inflammation, or both**



This figure shows the serum (also called soluble) transferrin receptor-ferritin index (sTfR-F index), the ratio of the serum TfR to the logarithm (base 10) of the serum ferritin in IDA (red circles), ACD/AI (green squares), and those with both conditions (IDA + ACD/AI; blue triangles). Median values are indicated by the horizontal bars. A value of ≤1.0 suggests ACD, while values ≥2 suggest IDA or IDA + ACD/AI.

IDA: iron deficiency anemia; ACD/AI: anemia of chronic disease/anemia of inflammation.

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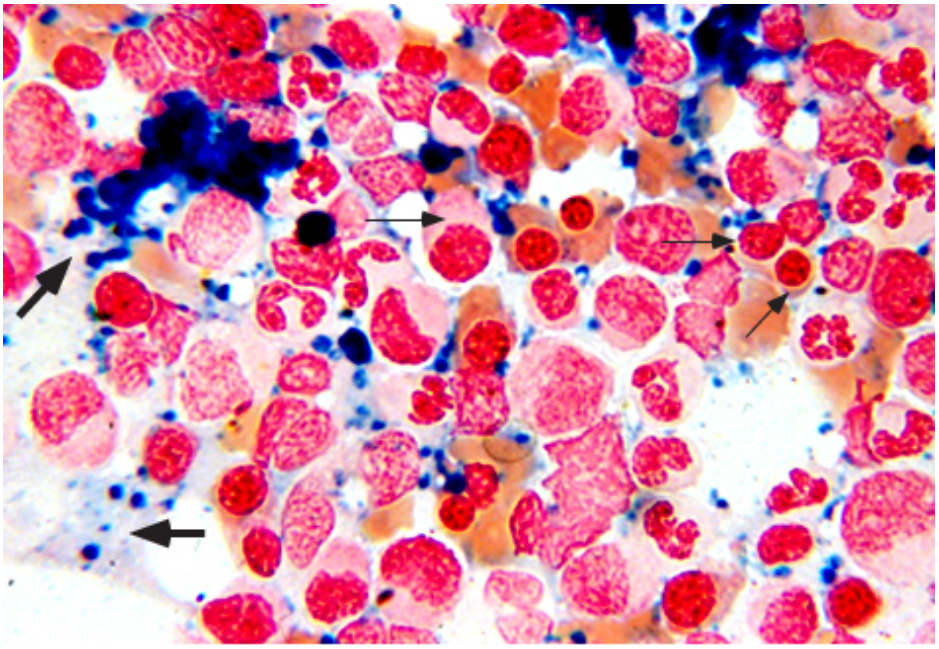
Graphic 79053 Version 3.0

Picture 1: **Anemia of chronic disease - bone marrow iron stain**

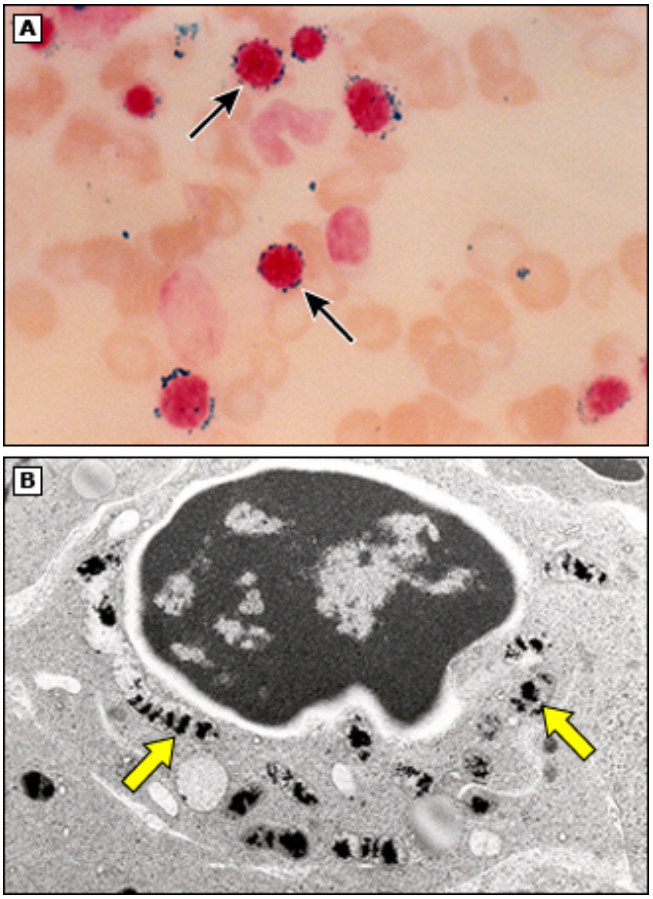


Section of a bone marrow aspirate taken from a patient with the anemia of chronic disease. The slide has been stained for iron (Prussian blue reaction) and counterstained with safranin to show nuclear detail. Note the increased iron staining within the voluminous cytoplasm of macrophages (thick black arrows), while there is no staining for iron within the cytoplasm of red blood cell precursors (thin black arrows). This pattern (abundant iron in macrophages and reduced to absent iron in red cell precursors) is quite typical for the anemia of chronic disease, and contrasts with iron deficiency, in which iron is absent from both macrophages and red cell precursors, while normal subjects demonstrate iron in macrophages and red cell precursors.

Slide provided by Stanley L Schrier, MD.

Graphic 70226 Version 2.0

Picture 2: **Bone marrow – ring sideroblasts**



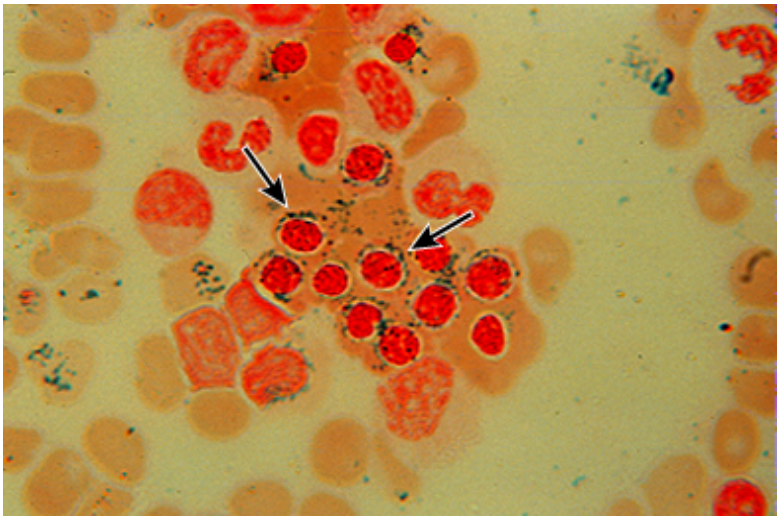
(A) Bone marrow aspirate stained with Prussian blue to show red cell precursors with numerous iron-positive granules surrounding the nucleus, in some cases forming a complete ring (arrows).

(B) Electron micrograph of a single erythroblast with iron-laden mitochondria (electron-dense deposits and thick arrows) clustered near its nucleus.

Courtesy of Sylvia Bottomley, MD.

Graphic 62536 Version 4.0

Picture 3: **Ring sideroblasts in refractory anemia with ring sideroblasts (RARS)**



Prussian blue stain of the bone marrow in a patient with refractory anemia and ring sideroblasts (RARS). Blue-stained ferritin iron deposits in the mitochondria of erythroid precursors form an apparent ring around the nucleus (arrows).

Courtesy of Stanley L Schrier, MD.

Graphic 65887 Version 6.0

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